

# Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C<sub>60</sub> Graphullerene

Xingqiang Chen and Qixing Wang

*Department of Physics*

*Xiamen University*

*Xiamen 361005, China*

(Dated: June 21, 2026)

Under a rigid-strain upper-bound protocol with fixed fractional coordinates, reported periodic  $|\mathcal{S}|$  magnitudes bound—rather than predict—equilibrium total-energy cross terms until sign-qualitative ionic-relaxation and alternate-placement benchmarks complete (Tables II and III; Table II 2/4 fixed-cell corners converged). Separate strain-only and doping-only workflows remain common, but total-energy effects under joint load are not necessarily additive. In quasi-hexagonal C<sub>60</sub> graphullerene we quantify the omitted cross term with the synergy order parameter  $\mathcal{S}$  (meV per atom) [Eq. (1)]. On tetramers ( $\sim 5\%$  nominal doping) and fifteen periodic  $n \times \text{C}_{60}$  supercells ( $n = 1\text{--}8$ , B/N/P @  $+3\%$ , one substituent per C<sub>60</sub>,  $\sim 1.67$  at.%), tetramer  $\alpha$  from the archived legacy PBE rigid grid (Table IV) is opposite for N ( $-303$  meV/%) and B ( $+57$  meV/%), whereas periodic PBE+D3 yields  $|\mathcal{S}|$  up to  $31.9$  meV/atom at fixed coordinates for P at  $n=1$  and  $+3\%$  strain, with P showing the largest  $|\mathcal{S}|$  despite the smallest tetramer  $|\alpha|$ . Sequential high-throughput screens can introduce systematic errors in the estimation of *relative* stability margins when local dopant distortion couples nonlinearly to external strain. Marcus reorganization energies and nearest-neighbor hopping parameters (Figs. 3 and 4) are reported for transport context but do not affect the non-additivity conclusions.

## I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate  $E_{\text{tot}}(\epsilon, \delta)$  from *separate* doping-only and strain-only calculations [1–3]. Synthetic carbon allotropes with extended  $\pi$  conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a periodic total-energy cross term under joint biaxial  $(\epsilon, \delta)$  load (fifteen-point periodic audits here use PBE+D3; tetramer  $\alpha$  in Table IV uses a separate legacy PBE archive). Quasi-hexagonal (qHP) C<sub>60</sub> graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C<sub>60</sub> monolayers [4]; recent surveys of covalent fullerene networks emphasize synthesis, catalysis, and transport properties [8] but do not report a periodic total-energy cross term under joint biaxial  $(\epsilon, \delta)$  load. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [9], so external biaxial load couples to an already distorted cage baseline. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 10, 11], while substitutional B/N engineering can yield spin-polarized half-semiconductor gaps [12], and related 2D fullerene networks show anisotropic mechanical and tribological response under load [13, 14] and endohedral qHP C<sub>60</sub> cage filling reshapes band gaps [15]; none of these reports a periodic total-energy cross term under joint biaxial  $(\epsilon, \delta)$  load, motivating explicit coupled grids rather than an additive superposition of composition and strain effects. Non-additive coupling is documented in other

two-dimensional networks [1, 16]; a quantitative *total-energy* synergy order parameter  $\mathcal{S}$  [Eq. (1)] has not been reported for substitutional B/N/P qHP C<sub>60</sub> tetramers and  $n = 1\text{--}8$  supercells; here, fifteen periodic  $(n, \delta)$  points at  $+3\%$  strain use PBE+D3, whereas tetramer  $\alpha$  in Table IV comes from the archived legacy PBE rigid grid (no D3).

We address three questions on this model network: (1) how opposite linear strain coefficients  $\alpha$  (meV per 1% biaxial strain of the *total* cell energy) emerge for donor vs. acceptor dopants; (2) how large the omitted synergy cross term  $|\mathcal{S}|$  (meV/atom) is when strain and substitution are applied jointly; and (3) whether  $|\mathcal{S}|$  tracks  $|\alpha|$  or instead reflects local structural distortion. Using  $\gtrsim 90$  converged CP2K tasks (Table I), we define  $\mathcal{S}$  on periodic PBE+D3 grids (fifteen  $(n, \delta)$  points at  $+3\%$  strain) and report opposite donor/acceptor  $\alpha$  on the archived tetramer strain grid (Table IV), demonstrating that the largest periodic  $|\mathcal{S}|$  at fixed coordinates (up to  $31.9$  meV/atom for P at  $n=1$ ) does not coincide with the smallest tetramer  $|\alpha|$  for size-mismatched P. *Local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy:  $|\mathcal{S}|$  can exceed what linear  $\alpha$  alone would suggest, so *relative* stability margins from  $\epsilon = 0$  substitution energies or strain-only elastic increments need not survive a coupled load even if they do not reorder dopants with widely separated  $E_{\text{sub}}$  values. Reported  $|\mathcal{S}|$  magnitudes are protocol upper bounds at fixed coordinates until ionic-relaxation and alternate-placement benchmarks complete (Tables II and III; Table II 2/4 fixed-cell corners converged). Marcus  $\lambda$  and hopping observables (Figs. 3 and 4) characterize transport-relevant quantities but are not used to infer  $\mathcal{S}$ .

## II. COMPUTATIONAL METHODS

### A. Electronic-structure setup

All periodic audit and charged-state tasks use CP2K [17] with PBE [18] and Grimme DFT-D3 (Becke–Johnson damping) [19]; tetramer  $\alpha$  and  $E_{\text{sub}}$  in Table IV come from the archived legacy PBE rigid grid (no D3). Table I summarizes basis sets, SCF convergence criteria, and the full task inventory. We do not employ the Koopmans-compliant rVV10 approach used in Ref. 4, and instead adopt a PBE+D3 protocol for periodic and charged-state tasks suitable for large-scale parameter scanning, at the cost of absolute accuracy in polaron binding energies.

### B. Notation and conventions

Biaxial strain is denoted  $\epsilon$  (percent); dopant species  $\delta \in \{B, N, P\}$ . The synergy order parameter is denoted by the calligraphic symbol  $\mathcal{S}$ , with units of meV per atom [Eq. (1)].  $\pi$ -projected densities of states are written  $\pi$ -DOS (never bare “pi” in equations).

**Rigid-strain protocol.** Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported  $\alpha$  and  $\mathcal{S}$  under this protocol are *upper bounds* at fixed coordinates, not equilibrium thermodynamic quantities. Table II benchmarks four fixed-cell geometry-optimization corners (pristine and  $P$ -substituted tetramer at  $\epsilon=0$  and  $+3\%$ ) to test whether  $\mathcal{S}$  retains *sign* after ionic relaxation *within the same tetramer grid* (magnitude ratios are not quantitative until a matched-functional rigid reference exists); it does *not* validate periodic  $n=4$  magnitudes, which differ in dopant concentration and boundary conditions. Main-text  $|\mathcal{S}|$  values therefore bound—rather than predict—meV/atom-scale non-additivity until Tables II and III benchmarks complete (Sec. IID3, Sec. IID2).

**Plane-wave cutoff (size scaling).** Archived CP2K inputs use 400 Ry for  $n \leq 4$  and 350 Ry for  $n \geq 6$  (Table I); comparisons at fixed  $n$  always share the same cutoff. Cross- $n$  trends therefore mix a modest plane-wave cutoff change at  $n = 6$ ; we do *not* interpret the reported  $N \mathcal{S}$  sign change between  $n = 6$  and  $n = 8$  as a physical size effect until the matched 400 Ry  $n=6$  control in Table I completes.

### C. Substitutional doping and grids

Tetramer models ( $4 \times C_{60}$ , 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates); these dopant configurations are not thermodynamically optimized ground states. A second tetramer placement ensemble (18 single-point energy tasks; Table III) tests whether tetramer  $\alpha$  and  $\mathcal{S}$  are placement-

sensitive at  $\sim 5\%$  nominal doping. All periodic supercells propagate the same dopant site map (one substituent per  $C_{60}$ ); quantitative  $|\mathcal{S}|$  on the fifteen-point grid is therefore configuration-specific, whereas donor/acceptor sign trends for  $\alpha$  on the reference tetramer placement are discussed as element-intrinsic qualitative expectations pending the alternate seed-137 map in Table III (Sec. IID2). Periodic scaling cells place one substituent per  $C_{60}$  ( $n$  dopants among  $60n$  lattice atoms,  $\sim 1.67$  at.% for all  $n$ ), contrasting with the tetramer grid ( $\sim 5\%$  nominal). Counting a single dopant in the full supercell (e.g.,  $1/240=0.42$  at.% for the tetramer) is *not* the concentration used here; atomic percent is always dopants per lattice atom ( $n$  in  $60n$ ). Tetramer  $\alpha$ , gaps, and tetramer IPR/ $J$  trends (Fig. 4) are therefore not directly interchangeable with periodic  $\mathcal{S}(n)$  without a matched-concentration benchmark; all  $\mathcal{S}$  claims use the periodic protocol. In total,  $\gtrsim 90$  converged neutral tasks support the protocol-audited claims below (Table I); rigid-strain  $|\mathcal{S}|$  magnitudes are upper bounds until Tables II and III complete. Charged-state geometry optimizations (UKS, multiplicity 2, EPS\_SCF =  $10^{-6} E_h$ ) supply adiabatic ionization potentials and electron affinities ( $E_{\text{opt}}^{\pm}$ ). Eight vertical single-point energy evaluations compute  $E_{\text{vert}}^{\pm}$  at the *neutral*-geometry coordinates with charged UKS settings matched to the relaxed state (EPS\_SCF =  $10^{-7} E_h$  by default; B  $q = -1$ :  $10^{-6} E_h$ ), giving  $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$  following Ref. 4. These Marcus diagnostics appear only in Fig. 3 and do not enter Eq. (1).

### D. Energy observables

Linear strain sensitivity  $\alpha = dE/d\epsilon$  (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the archived 24-point rigid tetramer grid (Table IV; legacy PBE without D3). Periodic  $\mathcal{S}$  and size-scaling energies use PBE+D3 throughout (Table I). Positive  $\alpha$  means tension *raises* total energy; negative  $\alpha$  means tension *lowers* it under the rigid-strain protocol. Substitution energy  $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$  is the total-energy difference per dopant site at fixed strain *without* elemental chemical-potential corrections; it is *not* a thermodynamic standard formation enthalpy ( $\sim$ eV-scale tabulated values) and is used only for relative ranking within this protocol. In sequential high-throughput screens one often estimates  $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$ ; the neglected term is precisely  $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows under the rigid upper-bound protocol, the magnitude of this additive omission is best gauged against a substitution-energy reference scale: at  $n = 4$  and  $+3\%$  strain, P has  $|\mathcal{S}| = 23.7$  meV/atom while the tetramer P per-atom scale is  $|E_{\text{sub}}|n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$  eV/atom ( $|E_{\text{sub}}|=15.8$  eV/dopant,  $n_{\text{dop}}=12$  on the 240-atom tetramer, Table IV)—i.e.  $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}} \sim 3\%$  as an *order-of-magnitude* illustration only, because tetramer ( $\sim 5\%$ ) and periodic  $n=4$  ( $\sim 1.67$  at.%) concen-

trations differ and Table IV is not on the same functional as periodic  $\mathcal{S}$ ; this ratio is not used as a core nonlinearity test.

**Non-additive coupling metric.** We quantify strain–doping cross terms with the synergy order parameter  $\mathcal{S}$ :

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where  $\epsilon$  is biaxial strain,  $\delta \in \{B, N, P\}$ , and  $\Delta E(\epsilon, \delta)$  is the energy per atom relative to the pristine, zero-strain reference at the same supercell size  $n \times C_{60}$ . The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives  $\mathcal{S} = 0$ ;  $\mathcal{S} < 0$  indicates cooperative stabilization beyond the two-scan sum, whereas  $\mathcal{S} > 0$  indicates cooperative destabilization.

When  $\alpha_\delta = dE/d\epsilon$  denotes the *total*-cell linear strain coefficient (meV per 1% strain) at fixed  $n$ , a first-order estimate of the per-atom cross term is  $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$  at small  $\epsilon$  under the rigid-strain protocol; tetramer and periodic  $n$  use different  $N_{\text{atoms}}$  and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

Periodic-cell  $\mathcal{S}$  values are extracted from converged CP2K total energies (Table I); the fifteen-point +3% strain audit accompanies the open data release. We report  $\mathcal{S}$  for total energy only; an analogous gap cross term  $\mathcal{S}_{\text{gap}}$  is not computed here—main-text panel (a) shows that gap slopes need not track  $|\mathcal{S}|$ .

**Localization and coupling (Fig. 4).** Following Capobianco *et al.* [4], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling  $J$  from tetramer dimer fragments. IPR is  $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$  on a Mulliken-projected orbital grid;  $J$  is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant  $J \simeq 75\text{--}81$  meV reported for pristine qHP networks [4], but they provide an internally consistent trend under coupled  $(\epsilon, \delta)$ .

**Polaron binding.** Marcus reorganization energies follow the vertical–adiabatic decomposition [4]:  $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ , with separate internal ( $\lambda_{\text{loc}}^\pm$ ) and environmental ( $\lambda_{\text{env}}^\pm$ ) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Fig. 3. Charged geometry optimizations supply adiabatic IP/EA; vertical single-points at the neutral geometry yield Marcus  $\lambda^\pm$  (Fig. 3).

### 1. DFT calculator settings

Periodic total-energy and electronic-structure tasks use CP2K [17] with PBE [18] and Grimme DFT-D3 (Becke–Johnson damping) [19]. Tetramer  $\alpha$  and  $E_{\text{sub}}$

in Table IV come from the archived rigid tetramer grid (legacy PBE without D3). Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). The plane-wave cutoff is 400 Ry for  $n \leq 4$  and 350 Ry for  $n \geq 6$  (Table I); the atom-centered basis (DZVP-MOLOPT-SR-GTH) is unchanged across  $n$ . Tetramer models ( $4 \times C_{60}$ , 240 atoms) use B/N/P substitution at nominal 5% (12–13 dopant sites from archived coordinates). Periodic supercells place one substituent per  $C_{60}$  unit ( $n = 1\text{--}8$ ), giving  $n$  dopants among  $60n$  lattice atoms ( $\sim 1.67$  at.% for all  $n$ ).

### 2. Alternate-placement benchmark (Table III)

The *reference* placement in Table III is the archived rigid tetramer map underlying Table IV (legacy PBE without D3). The *alternate* placement repeats the full B/N/P strain grid ( $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ ; 18 rigid single-point tasks) with an independently rerolled substitution map at the same nominal 5% doping level (seed 137). Reference-placement  $\mathcal{S}_{+3\%}^{\text{tet}}$  for B/N/P is tabulated in Table III and Sec. III C; periodic  $\mathcal{S}(n)$  is not inferred from either tetramer map. Alternate tasks use PBE+D3 rigid RUN\_TYPE ENERGY at 400 Ry with EPS\_SCF =  $10^{-6} E_h$ ; they are *not* on the same exchange-correlation treatment as the reference  $\alpha$  column, so only within-placement trends are compared once the grid converges.

### 3. Ionic-relaxation benchmark (Table II)

The four tetramer corners in Table II are pristine and  $P$ -substituted cells at  $\epsilon=0$  and +3% biaxial strain (the Eq. (1) grid for  $P$ ). They use RUN\_TYPE GEO\_OPT with BFGS (maximum 300 ionic steps) at prescribed in-plane cell vectors; only fractional coordinates relax while lattice vectors remain fixed. BFGS stops at 300 steps or when the maximum and RMS force components fall below  $4.5 \times 10^{-4}$  and  $3.0 \times 10^{-4}$  Ha/bohr, or when the maximum and RMS atomic displacements fall below  $3.0 \times 10^{-3}$  and  $1.5 \times 10^{-3}$  bohr, respectively (archived inputs `relax*.geo.inp`). These tasks use the same PBE+D3 settings as periodic audits: 400 Ry plane-wave cutoff, DZVP-MOLOPT-SR-GTH basis, and EPS\_SCF =  $10^{-6} E_h$  with OT-DIIS and up to 20 outer SCF cycles. Rigid reference energies in Table II remain the archived legacy-PBE (no D3) single-point grid; relaxed and rigid corners therefore differ in exchange-correlation treatment, and only sign retention of  $\mathcal{S}$  is interpreted.

#### 4. Marcus vertical step and post-processing

Charged-state geometry optimizations supply adiabatic ionization potentials and electron affinities ( $E_{\text{opt}}^{\pm}$ ). Eight vertical single-point energy evaluations compute  $E_{\text{vert}}^{\pm}$  at the *neutral*-geometry coordinates with the charged UKS settings of the relaxed state, following Ref. 4;  $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$  (Fig. 3). Figure 3 schematically separates adiabatic IP (slanted arrow between neutral and charged minima) from the neutral-geometry vertical single-points that define  $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ ; adiabatic EA is quoted as text on the B and P panels only.  $\pi$ -projected densities of states use a 0.05 eV Gaussian broadening aligned to the Fermi level (Fig. 2). Inverse participation ratios and hopping integrals  $J$  are defined in Sec. II and Fig. 4; numerical values are listed in the Data Availability repository.

#### E. Validation benchmarks

The full calculation map (converged tasks, main-text usage, and pending controls) is in Table I; ionic-relaxation energies (Table II) and the alternate-placement grid (Table III) appear there when complete. Three validation benchmarks—rigid vs. relaxed tetramer  $\mathcal{S}_P$  (Table II), alternate dopant placement (Table III; reference-placement  $\mathcal{S}_{+3\%}^{\text{tet}}$  tabulated in Sec. III C; alternate grid PBE+D3 on seed 137, Sec. IID 2), and a matched 400 Ry  $n=6$  N control (Table I)—gate interpretation of the N  $\mathcal{S}(n)$  sign change cited in Secs. III C and IV. Table II has completed the pristine fixed-cell geometry-optimization corners at  $\epsilon=0$  and  $+3\%$  ( $E = -1363.9551$  and  $-1363.9550 E_h$ );  $\mathcal{S}_{\text{relaxed}}$  and sign retention await the two P-doped fixed-cell corners (Sec. IID 3).

Table IV and Sec. IIIB report  $E_{\text{sub}}$  using the substitution-energy definition above (no chemical-potential reference; not a formation enthalpy).

### III. RESULTS

We organize the results around total-energy non-additivity under the rigid-strain upper-bound protocol (Sec. IIB); Tables II and III gate relaxation- and placement-sensitive interpretations. Sec. III A summarizes band-edge context; Sec. III B reports linear strain coefficients,  $E_{\text{sub}}$ , and geometric strain response (Table V); Sec. III C gives  $\mathcal{S}(n)$  at  $+3\%$  strain across dopants and supercell sizes (Table I; fifteen-point periodic audit); Sec. IIID reports Marcus  $\lambda$  and  $\text{IPR}/J$  after the  $\mathcal{S}$  audit.

#### A. Electronic structure under strain and doping

Figure 1(a,b) summarizes tetramer HOMO–LUMO gaps (archived tetramer electronic-structure set) and

representative  $\pi$ -DOS (B @  $\epsilon=0$ ; color scheme follows Ref. 4). At  $\epsilon=0$ , B is near a closed  $\pi$  gap ( $E_g \approx 0.03$  eV), N doping pins the Fermi level inside the conduction band ( $E_g \approx -0.14$  eV), and P remains gapped ( $E_g \approx 0.05$  eV); tensile strain narrows the pristine gap ( $\sim 23$  meV/% on the three available strain points in the dataset). Donor vs. acceptor ordering of the  $\epsilon=0$  gaps tracks the opposite  $\alpha$  signs in Sec. IIIB, but band-edge slopes are not used to infer  $\mathcal{S}$ .

#### B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ( $|\alpha| \approx 1$  meV/%). On the tetramer grid (Table IV), substitution opens an order-of-magnitude spread in  $\alpha$ : N ( $-303.2$  meV/%, tension stabilizes the doped cell), B ( $+56.9$  meV/%, tension destabilizes), and P ( $-19.1$  meV/%, weakly stabilizing). The opposite sign for N vs. B persists across the  $\epsilon \in \{-5, \dots, +5\}\%$  grid at this fixed placement and is not an artifact of a single strain point; Table III will test whether the slopes are placement-sensitive. At  $\epsilon = +5\%$ , N remains the most tension-stabilized dopant ( $\alpha$  most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative  $\alpha$ , foreshadowing why its linear coefficient poorly predicts  $|\mathcal{S}| = 23.7$  meV/atom for P at  $n=4$  under  $+3\%$  strain (Sec. III C). Substitution energies at  $\epsilon = 0$  span  $E_{\text{sub}} = +73.4$  eV/dopant (B),  $-115.3$  eV/dopant (N), and  $-15.8$  eV/dopant (P)—raw total-energy differences without chemical-potential reference, not standard formation enthalpies—with  $n_{\text{dop}} = 12\text{--}13$  counted from input coordinates (Table IV). Figure 1(c) highlights the  $\alpha$  split; ranking dopants solely by  $|E_{\text{sub}}|$  at  $\epsilon=0$  is misleading once  $+3\%$  strain is applied. Table V links this mechanical spread to local geometry: P shows negligible strain-induced change in mean nearest C distance ( $\Delta \bar{d} \approx 3 \times 10^{-5}$  Å vs.  $\sim 8.5 \times 10^{-3}$  Å for B at  $+3\%$  load), foreshadowing the  $\alpha$ – $\mathcal{S}$  decoupling quantified in Sec. III C (Discussion mechanistic synthesis).

#### C. Non-additive strain–doping coupling

Figure 1(d) tracks protocol  $\mathcal{S}(n)$  (rigid-strain upper bounds) with one dopant per  $\text{C}_{60}$  at  $+3\%$  biaxial strain. At  $n = 4$  under the fixed periodic dopant map (reference tetramer placement in Table III; alternate seed-137 grid pending, Sec. IID 2),  $\mathcal{S} = -6.2$  meV/atom (B),  $+3.0$  meV/atom (N), and  $-23.7$  meV/atom (P); P shows the largest magnitude despite the smallest  $|\alpha|$  on the tetramer grid (Table IV). Across fifteen periodic points ( $n \in \{1, 2, 4, 6, 8\}$ , B/N/P),  $|\mathcal{S}|$  reaches 31.9 meV/atom for P at  $n = 1$  and remains  $\mathcal{O}(10)$  meV/atom at larger  $n$  for B and P, whereas N is smaller in magnitude at  $n = 4$  but comparable at  $n = 1$ . For N,  $\mathcal{S}$  changes sign between  $n = 6$  ( $+2.4$  meV/atom) and  $n = 8$  ( $-2.3$  meV/atom) at

TABLE I. Summary of converged DFT calculations and correspondence to main-text results. The  $\sim 90$  tasks cited in the main text are the converged calculations that directly support quantitative claims; the full archive includes additional preliminary grids (rows marked “not used”).

| Calculation set                 | Model                    | Converged tasks   | Main-text usage       | Notes                                                         |
|---------------------------------|--------------------------|-------------------|-----------------------|---------------------------------------------------------------|
| Early strain probe              | $2 \times C_{60}$        | 5                 | not used              | superseded by tetramer grids                                  |
| Single-molecule doping          | $C_{60}$                 | 11                | not used              | superseded                                                    |
| Early electronics               | $2 \times C_{60}$        | 24                | not used              | superseded by tetramer $\pi$ -DOS                             |
| Polaron IPR/ $J$ tetramer dimer | tetramer                 | 4                 | Fig. 4                | pristine/B $\times \epsilon=0, +3\%$                          |
| Tetramer strain energies        | tetramer                 | 24                | Table IV, $\alpha$    | legacy PBE rigid; no D3                                       |
| Optimal-design grid             | tetramer                 | 20                | not used              | archive only                                                  |
| Tetramer $\pi$ -DOS/gaps        | tetramer                 | 12                | Fig. 2; main panels   |                                                               |
| Geometry references             | tetramer relax           | 6                 | charged-state inputs  | reference geometries                                          |
| Charged polarons                | tetramer $q = \pm 1$     | 12 geometry opt.  | Fig. 3                | Marcus $\lambda^\pm$ and adiabatic IP/EA                      |
| Size scaling                    | $n \times C_{60}$        | 40 (+1 pending)   | main $\mathcal{S}(n)$ | 400 Ry ( $n \leq 4$ ), 350 Ry ( $n \geq 6$ ); cutoff400 contr |
| $n=6$ cutoff control            | $6 \times C_{60}$ N @+3% | pending           | N sign benchmark      | 400 Ry matched control (pending)                              |
| Ionic relaxation (tetramer)     | tetramer                 | 2/4 geometry opt. | Table II              | fixed-cell PBE+D3 GEO-OPT (BFGS; Sec. IID 3)                  |

TABLE II. Ionic-relaxation benchmark for the tetramer  $P$ -substitution synergy grid (four fixed-cell geometry-optimization corners: pristine and  $P$  at  $\epsilon=0$  and  $+3\%$ , same biaxial cell). This benchmark is *sign-qualitative only*: the rigid reference uses legacy PBE without D3, whereas relaxed corners use PBE+D3, so  $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$  is *not* a quantitative retention ratio until a matched-functional rigid grid is recomputed. We test whether  $\mathcal{S}$  retains *sign within the tetramer grid only*; it does *not* calibrate periodic main-text  $\mathcal{S}(n)$ , which uses different dopant concentration and boundary conditions. Rigid reference energies: archived fixed-coordinate tetramer single-point grid (legacy PBE, no D3). Relaxed energies: PBE+D3 fixed-cell geometry optimization (400 Ry) from archived inputs in the open repository.  $\mathcal{S}$  follows Eq. (1) with  $N=240$  atoms.

| Quantity                                 | Rigid (legacy tetramer)                                                | Relaxed (geometry opt.) |
|------------------------------------------|------------------------------------------------------------------------|-------------------------|
| $E(\text{pristine}, \epsilon=0) (E_h)$   | -1363.5512                                                             | -1363.9551              |
| $E(\text{pristine}, \epsilon=3\%) (E_h)$ | -1363.5522                                                             | -1363.9550              |
| $E(P, \epsilon=0) (E_h)$                 | -1370.4965                                                             | pending                 |
| $E(P, \epsilon=3\%) (E_h)$               | -1370.4890                                                             | pending                 |
| $\mathcal{S}$ (meV/atom)                 | +0.96                                                                  | pending                 |
| Sign of $\mathcal{S}$ vs. rigid          | pending (2/4 geometry opt.; $P$ -doped fixed-cell corners in progress) |                         |

the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until the matched 400 Ry  $n=6$  control in Table I and Tables II and III complete (Sec. IID 3). Core size-scaling claims focus on  $n \leq 4$  at matched 400 Ry cutoffs; B and P show monotonic  $|\mathcal{S}|$  decrease from  $n = 1$  toward  $n = 4$ , whereas  $n \geq 6$  trends are reported for completeness only and mix the 350 Ry production cutoff. We report provisional  $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$  fits in the open data release but refrain from drawing conclusions about the asymptotic  $\mathcal{S}_\infty$  value until cutoffs are matched across  $n$ . All periodic  $\mathcal{S}$  values quoted in this subsection use the fixed dopant-site map and are tabulated in Table I (fifteen  $n \times$  dopant points at  $+3\%$  strain). On the reference tetramer placement, legacy PBE rigid  $\mathcal{S}_{+3\%}^{\text{tet}}$  from Eq. (1) is  $+3.8$ ,  $+5.7$ , and  $+1.0$  meV/atom for B, N, and P (Table III); the pending alternate-placement grid uses PBE+D3 rigid single-points on an independent site map (Sec. IID 2) and is not used to infer periodic  $\mathcal{S}(n)$ . Negative  $\mathcal{S}$  (B, P at  $n = 4$ ) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive  $\mathcal{S}$  (N at  $n = 4$ ) means sequential screens *overestimates* stability under load.

*a. Ionic-relaxation checkpoint (partial).* Table II benchmarks four fixed-cell geometry-optimization cor-

ners on the tetramer  $P$ -substitution grid (pristine and  $P$  at  $\epsilon=0$  and  $+3\%$ ); pristine corners at both strains have converged (2/4), and the two  $P$ -doped corners remain in progress. For the pristine tetramer, relaxed total energies at  $\epsilon=0$  and  $+3\%$  are  $-1363.9551$  and  $-1363.9550 E_h$  ( $\Delta E \approx 1.2 \times 10^{-4} E_h$ ), roughly an order of magnitude smaller than the archived rigid legacy-PBE nominal strain increment ( $\sim 10^{-3} E_h$ ). Because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3 (Sec. IID 3), only *sign* retention of  $\mathcal{S}$  is interpreted once all four corners complete.  $\mathcal{S}_{\text{relaxed}}$  and sign comparison to the rigid  $+0.96$  meV/atom entry await the two  $P$ -doped fixed-cell corners; Sec. IV links this interim pristine strain-energy shrinkage to the upper-bound reading of load-dependent margins.

## D. Polaron transport diagnostics

Figure 3 reports Marcus reorganization energies from twelve charged-state geometry optimizations and eight vertical single-points at the neutral geometry (Sec. IID 4). Columns compare the *same* Marcus workflow across dopants: the slanted arrow is adiabatic IP,

TABLE III. Alternate dopant-placement sensitivity (*tetramer only*, rigid strain). The *reference* placement defines main-text tetramer  $\alpha$  (Table IV) and the periodic supercell site map (one substituent per  $C_{60}$ ); its  $\alpha$  and  $\mathcal{S}_{+3\%}^{\text{tet}}$  use the archived legacy-PBE grid (no D3). The *alternate* placement repeats the full B/N/P strain grid ( $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ ; 18 rigid single-points) on an independent seed-137 site map with PBE+D3 (Sec. II D 2); pending rows test placement sensitivity at  $\sim 5\%$  nominal doping without mixing periodic boundary conditions.  $\mathcal{S}_{+3\%}^{\text{tet}}$  follows Eq. (1) on the archived rigid tetramer four-corner grid ( $N=240$ ). Periodic  $\mathcal{S}(n)$  is *not* tabulated here because concentration and boundary conditions differ; main-text periodic magnitudes are configuration-specific until alternate periodic placements are computed.

| Dopant | Placement | $\alpha$ (meV/%) | $\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom) | Status              |
|--------|-----------|------------------|----------------------------------------------|---------------------|
| B      | reference | +56.9            | +3.8                                         | verified (Table IV) |
| N      | reference | -303.2           | +5.7                                         | verified            |
| P      | reference | -19.1            | +1.0                                         | verified            |
| B      | alternate | pending          | pending                                      | pending             |
| N      | alternate | pending          | pending                                      | pending             |
| P      | alternate | pending          | pending                                      | pending             |

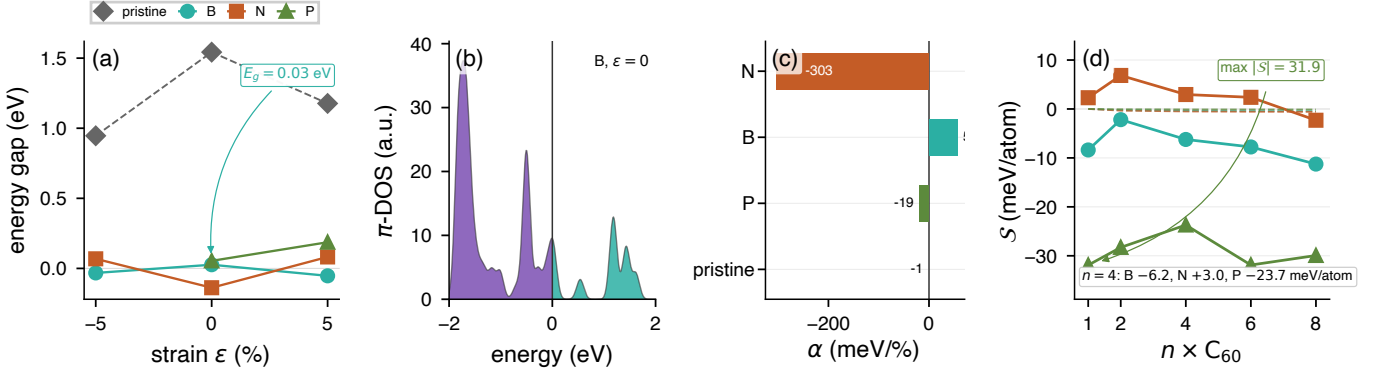


FIG. 1. Coupled  $(\epsilon, \delta)$  response in doped qHP  $C_{60}$  (PBE+D3). (a) HOMO-LUMO gap vs. strain ( $E_g$  callout: B @  $\epsilon=0$ ). (b)  $\pi$ -DOS for B @  $\epsilon=0$  (color scheme follows Ref. 4). (c) Mechanical  $\alpha$  (Table IV; legacy PBE rigid grid). (d)  $\mathcal{S}(n)$  at +3% (peak  $|\mathcal{S}|$  at P,  $n=1$ ; inset lists  $n=4$  values for B/N/P); Marcus  $\lambda$  and IPR/ $J$  in Figs. 3 and 4.

teal text gives adiabatic EA on B/P, and boxed  $\lambda^\pm$  values come from vertical minus adiabatic energies at the neutral geometry.

On the tetramer dimer grid, B substitution at  $\epsilon=0$  delocalizes the charged-state orbital (IPR  $75.0 \rightarrow 24.6$ ) while raising  $J$  from 26.6 to 46.4 meV; coupled B @ +3% gives  $J = 37.2$  meV and IPR 23.1. The criterion  $J > \lambda/2$  for polaron hopping dominance [4] is *not* met at PBE+D3 ( $J < \lambda/2$ ), so a polaron transition is not confirmed on this grid.

#### IV. DISCUSSION

**Context.** Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–15, 20, 21], the distinction here is a periodic *total-energy* audit of joint biaxial  $(\epsilon, \delta)$  load [Eq. (1)] on a rigid-strain upper-bound protocol (periodic PBE+D3; tetramer  $\alpha$  from legacy PBE, Table IV). Strain-engineering roadmaps for generic two-dimensional materials [3] likewise foreground band-edge and mobility levers under separate mechanical or chemical protocols rather than a matched four-corner total-

energy grid on covalent fullerene networks. Graphendofullerene derivatives report Curie temperatures tunable through *either* biaxial strain or electrostatic doping applied in isolation [11], reinforcing that joint  $(\epsilon, \delta)$  total-energy corners are not implicit in sequential protocol designs. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant-host size mismatch [21]; spin-resolved impurity engineering on qHP graphullerene instead targets half-semiconductor gaps [12]; isomer-engineered  $C_{70}$  network designs similarly tune band edges through topology choice rather than joint substitutional-mechanical load [22]; we instead quantify the omitted additive cross term  $\mathcal{S}$  in total energy. Quasi-harmonic thermal-expansion studies of qHP monolayers report anisotropic positive and negative lattice response tied to interfullerene bond topology [10, 20], whereas our audit holds fractional coordinates fixed at zero-temperature total energy (upper bound; partial ionic-relaxation and alternate-placement checkpoints in Secs. IIIC and IIE and Tables II and III). Strain-engineering studies on smaller fullerene graphullerenes instead emphasize magnetic or half-metallic transitions [23], which is orthogonal to the neutral total-energy audit pursued here. Band-

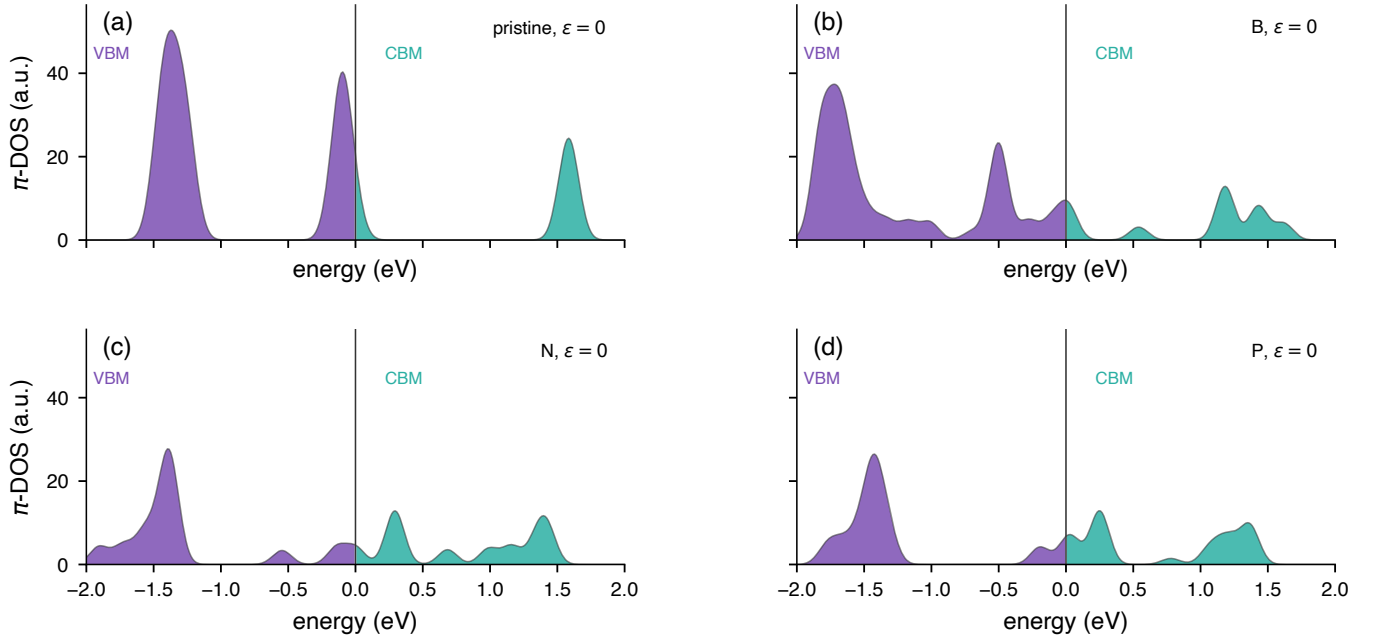


FIG. 2.  $\pi$ -projected density of states ( $\pi$ -DOS) for pristine and B/N/P-doped tetramers at  $\epsilon=0$  (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [4]).

TABLE IV. Linear strain sensitivities and substitution energies for tetramer models (archived rigid-strain grid, legacy PBE without D3; not mixed with periodic PBE+D3  $\mathcal{S}$  in Table I).  $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$  at  $\epsilon=0$  with no chemical-potential reference (not a standard formation enthalpy).  $n_{\text{dopants}}$  is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12).

| System   | $n_{\text{dop}}$ | $E(\epsilon=0)$ ( $E_h$ ) | $E(\epsilon=+5\%)$ ( $E_h$ ) | $\alpha$ (meV/%) | $E_{\text{sub}}$ (eV/dopant) |
|----------|------------------|---------------------------|------------------------------|------------------|------------------------------|
| Pristine | —                | −1363.5512                | −1363.5509                   | −1.0             | —                            |
| B-doped  | 13               | −1328.4900                | −1328.4090                   | +56.9            | +73.4                        |
| N-doped  | 12               | −1414.3812                | −1414.4053                   | −303.2           | −115.3                       |
| P-doped  | 12               | −1370.4965                | −1370.4888                   | −19.1            | −15.8                        |

edge, spin-resolved, and hopping metrics (Fig. 4) contextualize mechanism but are not used to infer  $\mathcal{S}$ .

**Mechanistic synthesis.** (i) **Opposite  $\alpha$  for N vs. B.** N introduces excess  $\pi$  electrons that strengthen bonding under tension (negative  $\alpha$ ), whereas B creates electron deficiency and antibonding  $\pi^*$  character that rises with lattice expansion (positive  $\alpha$ ). Quantitatively, the  $\epsilon=0$  gap ordering ( $E_g^N < E_g^B < E_g^P$  within PBE; Sec. III A) and dopant-resolved  $\pi$ -DOS in Fig. 1(b) and Fig. 2 align with donor vs. acceptor  $\pi$ -counting, while total-energy  $\alpha$  spans  $\sim 360$  meV/% between N and B on the tetramer grid (Table IV). The opposite N vs. B sign is reported for this fixed placement across  $\epsilon \in \{-5, \dots, +5\}\%$  (Table IV); Table III will repeat the full tetramer strain grid on an alternate seed-137 site map (PBE+D3 rigid; Sec. II D 2) to test slope sensitivity at  $\sim 5\%$  nominal doping without mixing periodic boundary conditions. Mayer bond order and Bader charge partitioning along the strain path are not yet computed (Limitations); periodic  $\mathcal{S}(n)$  magnitudes are likewise not averaged over alternate supercell placements until such

audits exist. (ii) **Large  $|\mathcal{S}|$  for P despite small  $|\alpha|$ .** P carries the largest covalent-radius excess (+30 pm vs. C; Table V) and pre-existing cage distortion at  $\epsilon=0$ , yet shows the smallest strain-induced change in mean nearest C distance ( $\Delta \bar{d} \approx 3 \times 10^{-5}$  Å vs.  $\sim 8.5 \times 10^{-3}$  Å for B/N) and the smallest site-to-site spread at +3% ( $\sigma(\bar{d}) \approx 4 \times 10^{-4}$  Å vs.  $\sim 2 \times 10^{-2}$  Å for B/N; Table V). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than to a uniform elastic response, yielding a cross term invisible in separate scans but dominant when both strain and substitution are applied simultaneously—consistent with P’s intermediate  $E_{\text{sub}}$  yet the largest  $|\mathcal{S}|$  at  $n=4$  (23.7 meV/atom; Table I) despite weak tetramer  $|\alpha|$ . Sec. III B presents this Table V geometric decoupling before the  $n=4$   $\mathcal{S}$  ranking in Sec. III C: tetramer  $|\alpha|$  ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the largest cross term. (iii) **N  $\mathcal{S}$  sign change with  $n$ .** With one substituent per  $C_{60}$ , electrostatic fields dilute at large  $n$ ; any N sign flip between  $n=6$  and  $n=8$  is confounded with the 400→350 Ry cutoff



TABLE V. Local dopant–cage structure from archived tetramer XYZ (mean nearest C distance  $\bar{d}$  in Å; standard deviation  $\sigma(\bar{d})$  across dopant sites; covalent-radius excess vs. C in pm). Strain-induced change  $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$  summarizes how tensile load couples to pre-existing distortion (Discussion mechanistic synthesis (ii),(vi)). Values are computed from archived tetramer XYZ coordinates in the open data release; Mayer bond order and Bader charges are not included.

| Dopant | $\bar{d}(\epsilon=0)$ (Å) | $\bar{d}(\epsilon = +3\%)$ (Å) | $\Delta\bar{d}$ (Å) | $\sigma(\bar{d})$ @ +3% (Å) | $\Delta r_{\text{cov}}$ (pm) |
|--------|---------------------------|--------------------------------|---------------------|-----------------------------|------------------------------|
| B      | 1.384                     | 1.393                          | +0.0085             | 0.019                       | +7                           |
| N      | 1.393                     | 1.402                          | +0.0087             | 0.024                       | −6                           |
| P      | 1.384                     | 1.385                          | +0.00003            | 0.0004                      | +30                          |

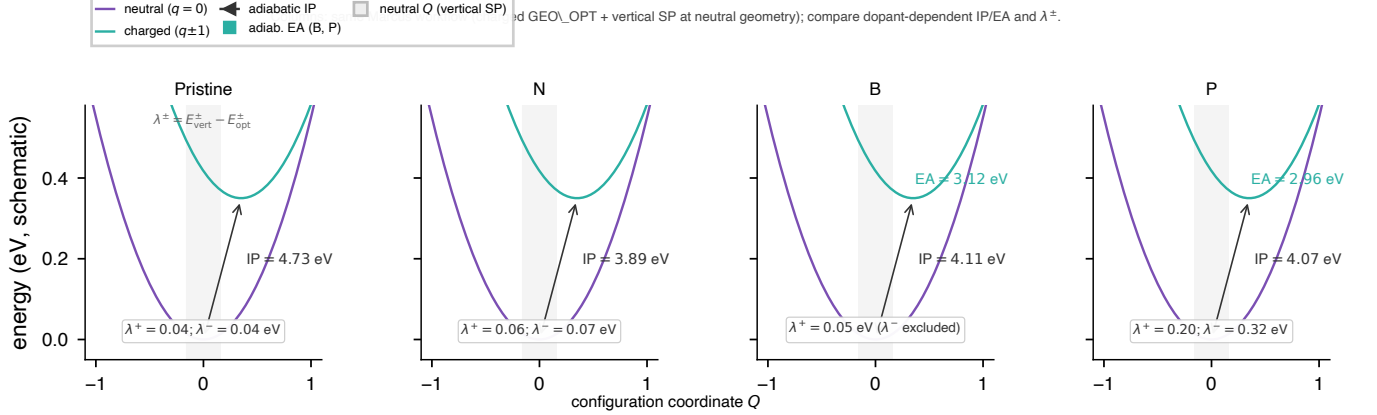


FIG. 3. Marcus reorganization energies across dopants (12/12 charged-state geometry optimizations; eight vertical single-points at the neutral geometry). Each column applies the same protocol; the slanted arrow marks adiabatic IP ( $q=+1$  GEO.OPT), teal labels give adiabatic EA where computed (B, P), and the shaded  $Q \approx 0$  band marks the neutral geometry used for vertical single-points ( $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ ). Pristine  $\lambda^\pm = 0.041/0.042$  eV; N  $\lambda^\pm = 0.058/0.065$  eV; B  $\lambda^+ = 0.051$  eV; P  $\lambda^\pm = 0.200/0.315$  eV. The electron-channel  $\lambda^-$  for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

step and remains uninterpreted (Sec. II E). Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of  $\alpha$ , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large  $n$ ). (iv) **Geometric vs. electronic nonlinearity.** For P, Table V shows that rigid tensile load barely moves the mean dopant–C distance ( $\Delta\bar{d} \approx 3 \times 10^{-5}$  Å) yet  $\mathcal{S}$  remains the largest at  $n=4$  (23.7 meV/atom), whereas B/N exhibit  $\sim 10^2 \times$  larger  $\Delta\bar{d}$  at +3% but smaller  $|\mathcal{S}|$  (6.2 and 3.0 meV/atom). At  $n=4$  the coupled correction for P exceeds the rigid per-atom strain increment from tetramer  $\alpha$  by two orders of magnitude ( $|\mathcal{S}| \gg \alpha \times 3\%/N_{\text{atoms}}$ ; Methods energy observables), even though  $|\alpha|$  ranks P as the weakest strain responder on the tetramer grid. (v) **Argument scope (concentration and boundary conditions).** We do *not* validate  $\mathcal{S}$  by extrapolating tetramer  $\alpha$  to periodic  $n=4$ : the  $\sim 5\%$  tetramer vs.  $\sim 1.67$  at.% periodic concentrations and tetramer vs. periodic boundary conditions are unmatched, so a first-order  $(\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$  estimate is excluded from the argument. Non-additivity is assessed only through Eq. (1) on matched four-corner grids at common  $(\epsilon, \delta)$ ; P in particular requires explicit coupled evaluation rather than sequential  $\alpha$  screening. (vi) **Decomposition outlook.** The Table V decoupling

indicates that the cross term is not reducible to uniform bond stretching alone; electronic rehybridization and pre-stored cage strain likely contribute, but their relative weights are not partitioned here without Mayer/Bader analysis along the strain path (Limitations).

(vii) **Significance versus DFT uncertainty.**  $\mathcal{S}$  is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. At  $n=4$  and +3% strain,  $|\mathcal{S}| = 23.7$  meV/atom for P is  $\sim 3\%$  of the tetramer P per-atom  $E_{\text{sub}}$  reference (Table IV) but still  $\gg$  rigid per-atom strain increments from  $\alpha$ . Large absolute uncertainties in PBE  $E_{\text{sub}}$  therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

**Design implications.** At  $n=4$  and +3% strain the coupled correction is  $|\mathcal{S}| = 23.7$  meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates differ by approximately a factor of four in the load-dependent *margin* even though widely separated  $E_{\text{sub}}$  ranks at  $\epsilon=0$  (e.g., B vs. P; Table IV) are unchanged. Dopants with larger covalent-radius excess (P: +30 pm vs. B: +7 pm; Table V) show the largest protocol  $|\mathcal{S}|$  under fixed coordinates; *size-mismatched* substituents may warrant coupled  $(\epsilon, \delta)$  evaluation rather than one-



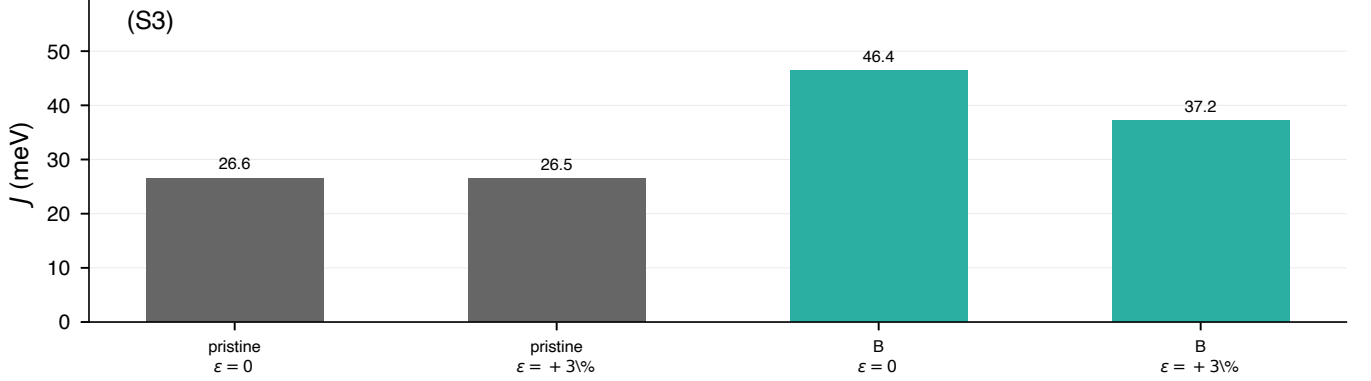


FIG. 4. Nearest-neighbor electronic coupling  $J$  on the tetramer dimer grid (pristine and B-doped cells at  $\epsilon=0$  and  $+3\%$ ; PBE+D3).

parameter pipelines once Tables II and III benchmarks complete. Interim pristine corners on the tetramer  $P$ -substitution four-corner grid (Table II; pristine  $\epsilon=0$  and  $+3\%$  converged,  $P$ -doped corners in progress; Sec. III C) shrink the nominal tensile strain increment by roughly an order of magnitude relative to the rigid legacy reference, supporting the upper-bound reading of protocol  $|\mathcal{S}|$  on load-dependent margins before the  $P$ -doped corners complete. The fifteen-point periodic grid tabulates protocol  $|\mathcal{S}|$  at each  $(n, \delta)$  as an audit of additive omission under the rigid protocol (Table I), not as quantitative design targets. At  $+3\%$  strain, P carries the largest additive-approximation error in protocol  $\mathcal{S}$  ( $|\mathcal{S}| = 23.7$ – $31.9$  meV/atom from  $n=4$  to  $n=1$  under fixed coordinates); qualitative co-design should treat P under tension as a *coupled* stability problem pending Tables II and III, not  $|E_{\text{sub}}|$  at  $\epsilon = 0$  alone. Where  $\mathcal{S} < 0$  (B and P at  $n=4$ ), coupled tension can stabilize doped cells beyond additive predictions; where  $\mathcal{S} > 0$  (N at  $n=4$ ), sequential screens overestimate stability under load. For chemistries with comparable  $|E_{\text{sub}}|$  or in multi-dopant competition, omitting  $\mathcal{S}$  can shift margins by several percent of the per-atom substitution scale—much larger than the rigid strain increment from  $\alpha$ —even when the global dopant rank at  $\epsilon=0$  is unchanged. Gap closing [Fig. 1(a)] does not track  $|\mathcal{S}|$ ; we do not compute  $\mathcal{S}_{\text{gap}}$ , so photonic or transport targets cannot be inferred from total-energy  $\mathcal{S}$  alone. For B at  $n = 4$ , both negative  $\mathcal{S}$  and positive  $\alpha$  imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating  $\mathcal{S}$ . For N at  $n=4$ , negative tetramer  $\alpha$  and positive  $\mathcal{S}$  present a second sign conflict: tension stabilizes the doped tetramer in isolation, yet the periodic coupled audit reports a positive cross term that sequential additive models fail to capture at  $\sim 1.67$  at.% doping. N  $\mathcal{S}$  vs.  $n$  is reported for completeness but excluded from design rules until Tables I–III controls converge.

**Validation protocol (Tables I–III).** Three valida-

tion benchmarks characterize protocol limits without altering the periodic magnitudes quoted above (Sec. III C reports the interim Table II pristine checkpoint). **Table II** runs fixed-cell geometry optimization on the four Eq. (1) corners for the tetramer  $P$ -substitution grid (pristine and  $P$  at  $\epsilon=0$  and  $+3\%$ ); because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3, the acceptance criterion is *sign retention* of  $\mathcal{S}$  after ionic relaxation within that tetramer grid—not a quantitative  $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$  ratio until a matched-functional rigid grid is recomputed. Interim pristine corners (Sec. III C) already show  $\Delta E \approx 1.2 \times 10^{-4} E_{\text{h}}$  after ionic relaxation versus  $\sim 10^{-3} E_{\text{h}}$  on the rigid legacy reference, so rigid-strain  $|\mathcal{S}|$  should be read as an upper bound on the strain increment until the  $P$ -doped corners complete; a matched-functional rigid grid would be needed before interpreting any magnitude ratio. It does *not* rescale periodic  $\mathcal{S}(n)$  because concentration and boundary conditions differ. **Table III** tests tetramer placement sensitivity: the reference map supplies archived legacy-PBE  $\alpha$  and  $\mathcal{S}_{+3\%}^{\text{tet}}$  (Table IV); the pending alternate map repeats the strain grid with PBE+D3 rigid single-points on an independent seed-137 site map (Sec. IID 2), so only within-placement trends are compared once converged. Reference-placement  $\mathcal{S}_{+3\%}^{\text{tet}}$  is already tabulated for B, N, and P ( $+3.8$ ,  $+5.7$ , and  $+1.0$  meV/atom; Sec. III C); these tetramer-only magnitudes are *not* extrapolated to periodic  $\mathcal{S}(n)$  because concentration and boundary conditions differ. Alternate-placement repeats will test whether  $\alpha$  slopes and  $\mathcal{S}_{+3\%}^{\text{tet}}$  signs are placement-sensitive at  $\sim 5\%$  nominal doping without direct cross-functional comparison to the reference column; qualitative claims about opposite N/B  $\alpha$  signs remain configuration-specific until that grid converges. **Table I** (cutoff400 row) isolates the  $400 \rightarrow 350$  Ry confound for  $6 \times C_{60}$  N @  $+3\%$  before any  $n \geq 6$  design statement is restored.

**Polaron transport.** The Marcus workflow in Figs. 3 and 4 parallels Ref. 4—charged-state geometry optimizations ( $E_{\text{opt}}^{\pm}$ ) plus vertical single-points at the neutral ge-

ometry ( $E_{\text{vert}}^{\pm}$ )—at the same PBE+D3 level as the periodic  $\mathcal{S}$  audit (tetramer  $\alpha$  in Table IV remains legacy PBE). Recent work shows that charged-supercell and neutral-defect semilocal routes to polaron self-interaction correction are formally equivalent when the polaronic structure is held fixed [24]; our Marcus  $\lambda$  protocol instead follows the charged vertical/adiabatic split of Ref. 4 without piecewise-linear or  $\gamma/\mu_{\text{DFT}}$  tuning, and remains orthogonal to the  $\mathcal{S}$  audit. Where that study linked hopping  $J$  and frontier-orbital localization to mobilities in pristine qHP networks, we retain the same observables for comparability (Fig. 4) but do *not* extend their mobility-centric ranking to neutral total-energy margins. Endohedral band-gap tuning [15] and substitutional  $\mathcal{S}$  address a second orthogonal knob (cage filling vs. cage substitution) and are not interchangeable for sequential-screen corrections.

**Limitations and outlook.** Three protocol limits are actively benchmarked (Table I; Sec. IIE): rigid strain (Table II), tetramer alternate-placement sensitivity (Table III), and mixed 400/350 Ry cutoffs across  $n$  (Table I cutoff400 control). All periodic  $\mathcal{S}(n)$  use a single fixed dopant configuration; magnitudes are not averaged over placement ensembles, and periodic alternate-site audits are not yet computed. The pending alternate tetramer grid (Table III) additionally differs in exchange-correlation treatment from the reference archive (PBE+D3 vs. legacy PBE), so placement and functional confounds are not separated in this revision. Additional limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10;  $\mathcal{S}_{\text{gap}}$  not quantified; Mayer bond-order and Bader charge partitioning along the strain path are not computed (Table V reports geometric metrics only); the electron-channel reorganization energy  $\lambda^-$  for B doping is unphysical under the fixed neutral-geometry vertical protocol and is excluded from rate estimates; mobility tensors are not computed—polaron transport anisotropy in related fullerene networks is treated only comparatively [4, 6] (Figs. 3 and 4).

## V. CONCLUSIONS

Sequential strain-doping workflows in covalent molecular networks can omit a meV/atom-scale total-energy cross term when local dopant distortion couples nonlinearly to external biaxial load; under the rigid-strain protocol, quoted  $|\mathcal{S}|$  magnitudes bound—rather than predict—equilibrium cross terms until ionic-relaxation and alternate-placement benchmarks complete (Tables II and III; Table II 2/4 fixed-cell corners converged). In qHP  $\text{C}_{60}$  graphullerene we answered the three Introduc-

tion questions under periodic PBE+D3 audits (tetramer  $\alpha$  from the legacy PBE archive in Table IV): **(1)** opposite donor/acceptor  $\alpha$  for N vs. B on the archived tetramer grid (Table IV); **(2)** meV/atom-scale omitted cross terms on a fifteen-point periodic grid ( $|\mathcal{S}|$  up to 31.9 meV/atom for P at  $n=1$  at fixed coordinates; Table I); **(3)**  $|\mathcal{S}|$  governed by local cage distortion (Table V) rather than tetramer  $|\alpha|$  for size-mismatched P. Ionic-relaxation, alternate-placement, and matched-cutoff controls (Tables I–III) will test whether these rigid-strain upper bounds and configuration-specific magnitudes survive relaxation and placement averaging; on the Table II four-corner grid, converged pristine corners already indicate a much smaller strain increment after fixed-cell ionic relaxation than on the rigid legacy reference (Sec. III C), so  $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$  is assessed for *sign* retention only until the  $P$ -doped corners complete and a matched-functional rigid grid exists. Coupled  $(\epsilon, \delta)$  screening is advisable for size-mismatched substituents and close substitution-energy competitors, complementing strain-only mechanical [13], derivative-network [11], and endohedral band-edge [15] axes. Future work completes the Table II  $P$ -doped ionic-relaxation corners and a matched-functional rigid grid for quantitative retention ratios; on Table III, completes the seed-137 alternate-placement PBE+D3 grid and adds PBE+D3 rigid repeats on the reference placement so placement and exchange-correlation confounds are not conflated (reference-placement  $\mathcal{S}_{+3\%}^{\text{tet}}$  already tabulated on the legacy archive). Extends placement tests to periodic cells, adds Mayer/Bader strain-path partitioning, and explores polaron-aware semilocal corrections beyond the Marcus vertical protocol.

## ACKNOWLEDGMENTS

We acknowledge computational resources and support from Xiamen University.

## DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including converged computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text. Pending ionic-relaxation, alternate-placement, and matched-cutoff validation tasks are documented in Tables I–III. A tagged release matching the manuscript revision will be issued upon acceptance.

[1] S. Chen, H. Liu, and K. Wang, untangling the respective effects of heteroatom doping and strain on elec-

tronic transport in 2D materials, *Advanced Materials* **36**,

- 2301234 (2024).
- [2] H. Li, M. A. Shakoori, and R. Wang, strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
  - [3] A. K. Katiyar and J. H. Ahn, strain-engineered 2D materials: challenges, opportunities, and future perspectives, *Small Methods* **9**, 2401404 (2025).
  - [4] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).
  - [5] Y. Yang, W. Zhang, H. Liu, and K. Li, two-dimensional carbon allotropes with tunable direct band gaps and high carrier mobility, *Appl. Surf. Sci.* **536**, 147815 (2021).
  - [6] P. Silva, R. M. Faria, L. G. Ferreira, and J. L. F. Fatima, large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
  - [7] Y.-B. Shi, S.-H. Lv, Z.-F. Shao, H.-K. Dong, S. Cao, and P. Qian, a first-principles study of 1d and 2d C<sub>60</sub> nanostructures: strain effects on band alignments and carrier mobility, *J. Phys.: Condens. Matter* **35**, 225701 (2023).
  - [8] R. Lv, J. Xi, M. Chen, and S. Yang, Covalent fullerene networks: preparation, properties and catalytic applications, *Rev. Mater. Res.* **2**, 100195 (2026).
  - [9] G. Makov and U. Argaman, structure and properties of graphullerene: a semiconducting two-dimensional C<sub>60</sub> crystal, *npj Comput. Mater.* **9**, 211 (2023).
  - [10] B. Peng and M. Pizzochero, monolayer C<sub>60</sub> networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
  - [11] D. López-Alcalá, Z. Hu, and J. J. Baldoví, graphendo-fullerene: a novel molecular two-dimensional ferromagnet, *Chemical Science* **16**, 7659 (2025).
  - [12] M. A. Khan and M. N. Leuenberger, tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
  - [13] J. Wang, K. Li, and H. Zhang, simulation on stress-related anisotropy of qTP C<sub>60</sub> and qHP C<sub>60</sub>: implications for optoelectronic nanodevices, *ACS Applied Nano Materials* **7**, 3456 (2024).
  - [14] F. Qiu, H. Wang, P. Zhang, and Y. Li, atomic-scale insights into the mechanical and tribological properties of graphullerene, *Tribology International* **204**, 110756 (2025).
  - [15] M. L. P. Junior, R. M. Tromer, L. A. R. Junior, and D. S. Galvão, endohedral derivatives of the recently synthesized two-dimensional fullerene networks: electronic and optical insights from first-principles calculations, *arXiv preprint* (2026), arXiv:2603.10142 [cond-mat.mtrl-sci].
  - [16] M. Alihosseini and M. Neek-Amal, Strain tuning of optoelectronic properties of covalent organic framework bilayers and heterostructures, *Phys. Rev. B* **108**, 165403 (2023).
  - [17] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, Quickstep: Fast and accurate density functional calculations using a mixed gaussian and plane waves approach, *Computer Physics Communications* **167**, 103 (2005).
  - [18] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77**, 3865 (1996).
  - [19] S. Grimme, S. Ehrlich, and L. Goerigk, effect of the damping function in dispersion corrected density functional theory, *Journal of Computational Chemistry* **32**, 1456 (2011).
  - [20] A. Shaikh, J. Wu, and B. Peng, negative and positive anisotropic thermal expansion in 2D fullerene networks, *arXiv preprint* (2025), arXiv:2504.02037 [cond-mat.mtrl-sci].
  - [21] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashenninnikov, strain-modulated defect engineering of two-dimensional materials, *npj 2D Materials and Applications* **8**, 33 (2024).
  - [22] J. Xu, H. Li, Z.-X. Guo, Y.-K. Li, and G.-L. Hou, designing two-dimensional fullerene networks from C70: a stable and synthesizable platform for electronic band engineering, *J. Phys. Chem. Lett.* **16**, 8321 (2025).
  - [23] J. Nie, P.-Y. Liang, J.-X. Li, W.-X. Li, D.-X. Liang, F. Lin, and S.-Z. Chen, strain-induced magnetism and half-metallicity in monolayer graphullerenes of the smallest fullerene C<sub>20</sub>, *Phys. Rev. B* **113**, 045401 (2026).
  - [24] S. Falletta, J. Coulter, J. B. Varley, D. Åberg, B. Sadigh, B. Kozinsky, and A. Pasquarello, Equivalence of charged and neutral density functional formulations for correcting the many-body self-interaction of polarons, *arXiv* (2025), arXiv:2511.02159 [cond-mat.mtrl-sci].